

CHANGES made after technical discussion on 3rd July, 2026

****** All BOM changes have been highlighted in yellow

CHANGE 1: Added BSS138 n-type MOSFET to be used as a switch to break the connection for battery monitoring, with gate of BSS138 connected to PA10 (GPIO output mode)

We can configure the BSS138 to turn the switch on and off whenever needed

CHANGE 2: Buzzer driver circuit installed by adding a protection 330ohm resistor in series to the buzzer circuit in series and matched the clock channel on both driving pins of buzzer to be PA11(TIM1_CH1) and PB2 (TIM1_CH1N)

CHANGE 3: Using ARM-10 connector included in the STlink V3SET as i couldn't fit the STDC14 version on the board due to physical and mapping constraints

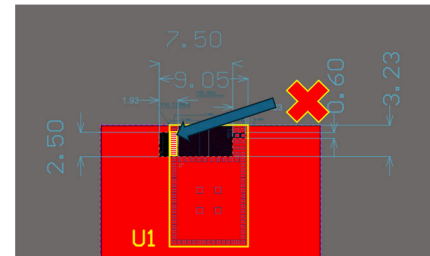
Table 15. ARM10-compatible connector pinout (target side)

Pin number	Description	Pin number	Description
1	T_VCC ⁽¹⁾	2	T_JTMS/T_SW ^{DI} O
3	GND	4	T_JCLK/T_SWCLK
5	GND	6	T_JTDO/T_SWC ⁽²⁾
7	T_JRCLK ⁽³⁾ /NC ⁽⁴⁾	8	T_JTDI/NC ⁽⁴⁾
9	GNDDetect ⁽⁵⁾	10	T_NRST

1. Input for STLINK-V3SET.
2. SWO is optional, required only for Serial Wire Viewer (SWV) trace.
3. Optional loopback of T_JCLK on the target side, required if loopback is removed on the STLINK-V3SET side.
4. NC means not required for the SWD connection.
5. Tied to GND by STLINK-V3SET firmware. It might be used by the target to detect the tool.

As a general rule, ground connections (marked GND) should be grounded. However, there are some exceptions: pads #2, #3, #4, #5, #56, and #60 should be left unconnected.

Figure 35. Error 1: do not connect NC pads on the left



CHANGE 4: Followed AN6305 to leave pins 2, 3, 4, 5, 56 and 59 ungrounded and isolated and created a no power and signal plane ruled zone (Back copper [MCU layer] , In2 [3.3V power plane] and In1 [GND power plane]) under the desired placement as given in the above circuit.

CHANGE 5: Added 0 ohm resistor between ANT_in and ANT_out to use the internal antenna

BOM is updated at [BOM changed](#)

I have also updated the PCB constraints to match those on the website of PCBpower with 0 critical errors found on the board.

I have also marked all of the above changes on the schematic.